Insurance Claim Fraud Detection Using Machine Learning

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***Abstract*—Auto insurance claim fraud has become increasingly prevalent in recent years posing a significant challenge for the auto insurance industry. To address this issue there is a growing need for more effective strategies to combat auto insurance claim fraud. The objective of this research is to utilize machine learning techniques to develop an accurate model for detecting fraudulent auto insurance claims. By leveraging the power of machine learning, this research assists insurance professionals in determining whether an insurance claimant has engaged in fraudulent activities related to vehicle insurance. The effectiveness of classifier models K-Nearest Neighbors, Support Vector Machines and Random evaluated by their accuracy, precision, recall and F1 score. Python programming language was used to facilitate the implementation of machine learning algorithms and the development of user-friendly web applications. In conclusion, RF outperformed other models with the highest accuracy, precision, recall, and F1-score, contributing to improved fraud detection and providing insurance companies with an effective tool to reduce financial losses and uphold operational integrity.**

***Index Terms*—Auto Insurance Claim, Fraud Detection, Data Analytics, Machine Learning Techniques, Classification Models**

I. Introduction

Insurance operates on the principle of mutual benefit, providing protection against large and unforeseen losses [1]. In recent years, there has been a significant increase in public demand for insurance claims. However, insurance fraud poses a threat to this system [1]. Deliberate deceit aimed at insurance companies or agents for financial gain, insurance fraud has garnered significant attention from governments, societies, and businesses in recent years [2]. Auto insurance fraud accounts for approximately 80 percent of all insurance fraud cases [2]. It not only infringes upon the rights of ordinary insurance consumers but also disrupts the normal functioning of the auto insurance market and jeopardizes road safety. Therefore, it is the responsibility of all stakeholders to establish an effective insurance claims fraud detection system to mitigate the occurrence of such fraudulent cases. To develop a highly accurate fraud detection system, machine learning techniques can be applied. Machine learning has the potential to provide superior predictive accuracy and can be applied across the insurance value chain to identify risks, claims, and customer behaviors [3]. It offers a wide range of applications in insurance, including risk assessment, premium leakage detection, expense management, subrogation, process optimization, and fraud detection [3]. By leveraging machine learning techniques, a robust model for detecting fraudulent auto insurance claims can be constructed and integrated into insurance claim fraud detection systems, thereby reducing the number of fraudulent claims. There are numerous machine learning techniques available globally, each with its own strengths and suitability for different circumstances. The choice of machine learning technique depends on the characteristics of the dataset. As the developer possesses a labeled dataset, supervised learning classification techniques are appropriate for this project. Three classification techniques, namely SVM, Random Forests, and KNN, will be implemented in this research[41-42]. The aim of this research is to develop an accurate auto insurance claim fraud detection system using machine learning models and aid insurance companies in predicting potential insurance claim fraud and taking appropriate actions. By utilizing the power of machine learning algorithms and addressing the challenges of imbalanced datasets, the system will enhance fraud detection capabilities, reduce financial losses, and contribute to the overall stability and integrity of the insurance industry. .

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II. Literature Review

Auto insurance is one type that protects vehicle owners by providing coverage for accidents, accidental damage, or theft, thereby alleviating their financial burden [4]. While auto insurance offers numerous benefits, some individuals exploit flaws in the insurance claim system to commit insurance fraud. Fraudulent claims have been a longstanding challenge for insurance companies [5]. Notably, auto insurance fraud accounts for approximately 80 percent of insurance fraud cases [2]. Auto insurance fraud can take various forms, ranging from submitting fake documents to staging fictitious accidents or auto thefts [6]. Detecting these fraudulent claims poses a significant challenge for insurance companies, as the schemes become increasingly sophisticated. Auto insurance claim fraud leads to substantial losses for insurance companies each year, affecting not only them but also the individuals involved in car insurance [5]. In the United States, insurance claims fraud ranks as the second-largest white-collar crime, resulting in over 80 billion dollar in annual losses [7]. To offset these losses, insurance companies often pass on the costs to

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customers through higher insurance premiums [8], leading to increased financial burdens for policyholders. Recognizing the need to combat insurance fraud, some insurance companies have invested in the development of detection systems using machine learning techniques.

III. Methodology

Insurance companies can use some criteria to identify potential insurance claim fraud. According to the [9], the researchers have analyzed a real dataset of auto insurance fraud. The researchers have established criteria to identify insurance claim fraud from the dataset, including claim characteristics, car type, policy type, and demographic characteristics. Each criterion has several variables in the dataset:

Demographic characteristics:

- Accident Area: Urban regions seem to be where fraudulent claims predominate.
- Sex: Comparison with women, men commit fraud much more frequently.
- Year: Fraudulent claims are more likely to occur in the first two years than later.
- Age of Driver: Younger drivers (under 36 years old) are more likely to commit fraud.
- Address Change: Policyholders who recently changed their address are more likely to commit fraud.
- Fault: The policyholder is more likely to be dishonest and committed in fraudulent
- Number of Supplements: There are more likely to be no supplements in fraudulent claims.
- Police Report Filed: Most fraudulent claims do not report to police.

Vehicle Type:

- Make: The most fraudulent claims typically pertain to cars made by Honda, Toyota, and Pontiac.
- Age of Vehicle: People owning older vehicles (those with vehicles 5 years old or older) are more likely to commit fraudulent claims.
- Vehicle Category: Sedan vehicles are more likely to be involved in insurance claim fraud.
- Vehicle Price: Low value vehicles (under 30,000 dollar) are more likely involved in insurance claim fraud.

Policy Type:

- Base Policy: In contrast to liabilities, fraudulent claims are more likely to be of the Collision or All Perils type.
- Agent Type: It is more likely that the external agent will be involved in the fraudulent claim.

Claim Characteristics:

- Past number of Claims: Fraudulent claims tend to happen to people who have a history of 2 to 4 past claims.
- Deductible: Compared to claims with deductibles of any other denomination, fraudulent claims with deductibles of 400 dollar are much more prevalent.
- Week of Month Claimed: The middle of the month tends to be when the most fraudulent claims are filed.
- Month: The months which has the most accidents.
- Month Claimed: The months of January, May, October, and November are the ones where fraudulent claims are most likely to be made.

A. *Imbalanced Dataset*

Fig. 1 shows that a dataset that has a majority class, which is a class that has many more example distributions than other classes, is said to be imbalanced. This poses a challenge for training classifiers in data mining and classification problems, as many algorithms struggle to handle imbalanced class distributions. Imbalanced datasets commonly occur in various real-life scenarios such as spam detection, credit card fraud detection, and natural disaster detection [10]. In this research, the dataset is collected from "Angoss Knowledge Seeker" software, which has a dataset of auto insurance records in the United States from 1994 to 1996 [5]. It exhibits an imbalanced distribution, with a majority of non-fraud cases (14,497) and a minority of fraud cases (923). To overcome the imbalanced dataset problems, the developer can use multiple data sampling techniques to avoid bias in the data [11]. Under-

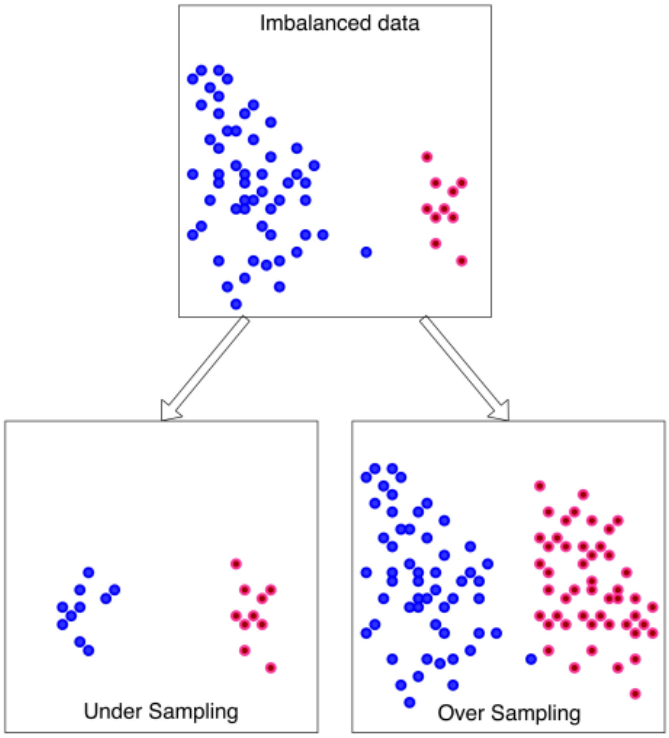

Fig. 1. Over-Sampling and Under-Sampling with Imbalanced Dataset [12]

sampling is removal of majority datasets either by using statistical techniques and randomly. Meanwhile, over-sampling is produced more datasets based on the minority class by using statistical techniques [13]. To deal with imbalanced data, under-sampling and over-sampling techniques are frequently utilized in ensemble learning or single model classifiers [12]. According to [5], the researchers encountered the issue of an imbalanced dataset in the collected data. To address this

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problem, they employed the Synthetic Minority Oversampling Technique (SMOTE), which is a popular data sampling technique [14]. SMOTE is an algorithm that tackles imbalance by generating synthetic observations based on existing minority observations, randomly selected from the k-nearest neighbors [15]. In this study, SMOTE was utilized to increase the number of instances in the imbalanced dataset from 15,420 to 16,343, resulting in the creation of a new balanced dataset. As the result of this study, the researchers applied the classification technique to the balanced dataset, with 93.72 percent accuracy for correct classification.

B. Machine Learning Techniques

Machine learning refers to a type of computer program that can learn on its own without requiring explicit programming by a person [16]. In modern times, the term "machine learning" is commonly used to describe a variety of programs used in big data analytics and data mining [16]. Unlike traditional programming, machine learning enables systems to automatically learn from data and adapt their behaviours[17]. Within the insurance industry, machine learning can be implemented to develop a system to detect fraudulent claim activities. To create a highly accurate insurance claims fraud detection system, machine learning techniques can be utilized. Machine learning has the potential to be applied throughout the insurance value chain, including risk identification, claims assessment, and analysis of customer behaviours, due to its superior predictive accuracy [3]. The applications of machine learning in insurance span various areas, such as assessing risk tolerance, preventing premium losses, managing expenses, handling subrogation, optimizing processes, and detecting fraudulent behaviours [3].

1) Support Vector Machine: Support Vectors are the closest data points or vectors to the hyperplane and have the greatest influence on where the hyperplane is located. Hyperplane is the best boundary in the SVM shown in Fig. 2. In the

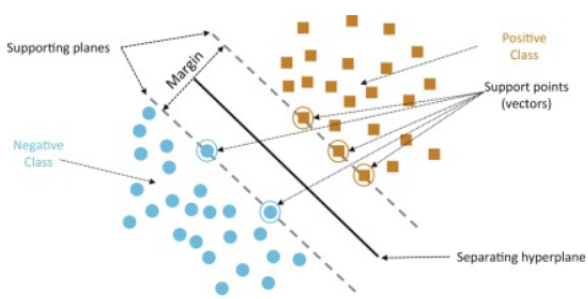

Fig. 2. Support Vector Machine [18]

field of fraud detection, Support Vector Machines (SVM) have been extensively studied to assist industries in reducing costs associated with fraudulent activities. For instance, SVM can learn to recognize fraudulent credit card transactions by analyzing numerous records of both fraudulent and legitimate credit card activity [19]. Similarly, in the case of insurance claim fraud, SVM can be employed to differentiate between fraudulent and non-fraudulent insurance claims.

2) K-Nearest Neighbor: K-Nearest Neighbors (KNN) is a type of non-parametric classification algorithm [20]. The new unlabeled data is classified by figuring out which classes its neighbors are a part of [21]. This idea applied in the calculation of the KNN algorithm, which allows us to classify tuples by fixing a particular value of K [21] Shown in Fig. 3. The

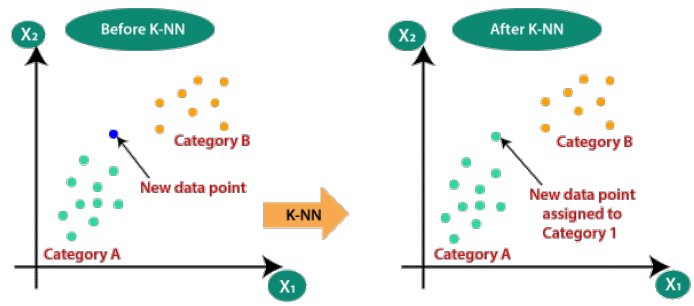

Fig. 3. K-Nearest Neighbour (KNN) [22]

K-Nearest Neighbour (KNN) algorithm has been extensively studied in the field of fraud detection to assist industries in reducing costs associated with fraudulent activities. In a study conducted by [23], the researchers utilized the KNN algorithm to develop a credit card fraud detection model. The findings of the study revealed that methods utilizing the KNN algorithm achieved an accuracy greater than 94.3 percent. On the other hand, the two methods with the lowest accuracy were not related to the KNN algorithm.

3) Random Forest: Random forest is a mixture of tree predictors where each tree depends on the value of a random vector [24]. In a random forest, each tree predictor is a decision tree. The data from the available datasets are randomly chosen to create each decision tree. By randomly choosing the characteristics of the decision tree, the correlation between trees will be reduced to increase accuracy and efficiency of the prediction model shown in Fig. 4.

Random Forest has been extensively studied in the field of fraud detection to aid industries in reducing costs associated with fraudulent activities. In a study conducted by [5], the researchers utilized the Random Forest Classification Technique to develop a procedure for identifying auto-fraud insurance cases.

C. Evaluation Metrics

1) Accuracy: Accuracy refers to the percentage of data classified accurately in a dataset and correctly predicted outputs. It is calculated by dividing the total number of observations used for classification by the number of observations that were correctly classified . Below is the accuracy formula shown in Fig. 5 :

2) Precision: Precision refers to the proportion of correctly identified outputs, or the exactness of the model. Below is the precision formula Fig. 6:

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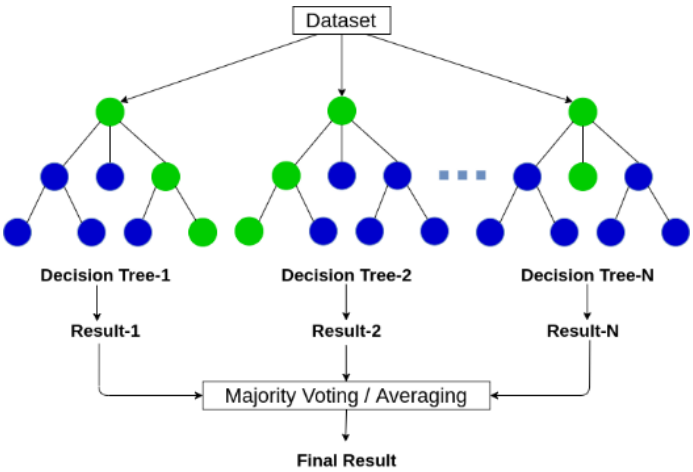

Fig. 4. Random Forest

$$Accuracy = \frac{TrueNegatives + TruePositive}{TruePositive + FalsePositive + TrueNegative + FalseNegative}$$

Fig. 5. Accuracy

3) Recall: Recall is the percentage of observations that belong to a given class that the model correctly classified. Below is the recall formula Fig. 7:

4) F1 Score: F1 score is the average of precision and recall. F1 considers both false positive and false negative, whereas precision only consider false positive and recall only consider false negative, respectively. When class distribution is uneven, F1 is highly helpful. Below is the F1 score formula Fig. 8:

IV. Result and Discussion

In this research, the researcher proposed SMOTE as an oversampling method to address the issue of class imbalance within the researcher's collected dataset.

Fig. 9 illustrates the distribution of the "FraudFound_P " variable, indicating a higher count of "0" compared to "1," suggesting a lower likelihood of fraud. The graph highlights an imbalanced class label. Conversely, Figures 10 illustrates the distribution of the "FraudFound_P" variable after applying the SMOTE technique. In contrast to Figure 9, the counts of "0" and "1" are now equal, indicating a 50 percent chance of fraud occurrence. Moreover, the graph demonstrates the successful resolution of the data imbalance issue. Once the imbalanced data issue was addressed, the researcher employed the Random Forest classification technique to detect fraud in auto insurance claims. In addition, other classification techniques, namely K-Nearest Neighbors and Support Vector Machines, were compared. Following the completion of development, the developers utilized evaluation metrics such as Accuracy, Precision, Recall, and F1 score to compare the performance of the three classifier models. The following table displays the comparison of the model's performance. The Table 1 shows the comparison of Machine Learning Models on Performance Metrics.

$$Precision = \frac{TruePositive}{TruePositive + FalsePositive}$$

Fig. 6. Precision

$$Recall = \frac{TruePositive}{TruePositive + FalseNegative}$$

Fig. 7. Recall

The experimental results demonstrate the superiority of the SMOTE with Random Forest approach in auto insurance claim fraud detection. The proposed method achieved the highest Accuracy, Precision, and F1 score compared to the alternative classification techniques, with 96.74percent, 98.72 percent, and 96.70 percent. The performance evaluation reveals that the SMOTE with Random Forests model outperforms both SMOTE with K-Nearest Neighbors and SMOTE with Support Vector Machines in identifying fraudulent claims, thus offering a more reliable and efficient solution to combat fraud in the auto insurance domain.

V. Conclusion

In conclusion, the development of an accurate auto insurance claim fraud detection system using machine learning techniques is crucial to address the increasing issue in insurance fraudulent activities. Insurance fraud, especially in auto insurance, not only impacts insurance companies but also affects the rights of consumers and road safety. By incorporating machine learning algorithms, a robust model can be constructed and integrated into insurance claim fraud detection systems to reduce the number of fraudulent claims. This research paper presents a comprehensive investigation into auto insurance claim fraud detection emphasizing the effectiveness of SMOTE combined with Random Forest. By

TABLE I
Comparison of Machine Learning Models on Performance Metrics

Metric	**SVM**	**Random Forest**	**KNN**	**Best Score**
Accuracy	84.31	96.74	93.14	Random Forest
Precision	81.03	98.72	91.05	Random Forest
Recall	89.90	94.76	95.79	K-Nearest Neighbour
F1 Score	85.24	96.70	93.36	Random Forest

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$$\text{F1 Score} = \frac{2 \times (\text{Precision} \times \text{Recall})}{\text{Precision} + \text{Recall}}$$

Fig. 8. F1 score [30]

Fig. 9. Distribution of the class label "FraudFound_P" without using SMOTE

conducting a comparative analysis with K-Nearest Neighbors and Support Vector Machines, the researche has demonstrated the superiority of the proposed approach. The research contributes to the advancement of fraud detection methodologies in the auto insurance industry offering insurance companies a powerful tool to mitigate financial losses and maintain the integrity of their operations. Future research directions may involve exploring additional data preprocessing techniques and incorporating other advanced or combination of multiple machine learning algorithms to further enhance fraud detection capabilities.

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